

Submission Instructions:

1. Write your Full Name, Student-ID, Theory Section on top of the first page
2. Your answer should be **handwritten**
3. Submit online here: <https://forms.gle/h84ifUQaUaAv4bWZ8>
4. **NO LATE SUBMISSION WILL BE ACCEPTED.**

Q1. [2+2 Marks] Refer to the table below to answer the following questions:

Movies				Actor		
MovieID	MovieName	MovieRating	Genre	ActorID	Actor	workedOn
M02	Mission Impossible	3.5	Action	TC1	Tom Cruise	M02
M12	It	3.1	Horror	FW3	Finn Wolfhard	M12
M14	Rush Hour	4.5	Comedy	JC5	Jackie Chan	M14

- Insert <"M12", null, 'X', "Thriller"> into the Movies table.
Identify all constraints which will be violated by this operation and **explain** how you will **solve** it.
- Insert <null, "Chris Evans", "M13"> into the Actor table.
Identify all constraints which will be violated by this operation and **explain** how you will **solve** it.

Q2. [8 Marks] You are tasked with designing a Driver Management System. The system must meet the following requirements:

- Drivers will have a unique DriverID, full name, email, and license number.
- Each vehicle has a unique VehicleID, model, capacity, and manufacturer.
- Each driver can operate many vehicles, but each vehicle can only be operated by one driver at a time.
- There are only 2 types of drivers: Full-time and Part-time. A driver can be both at the same time. Full-time drivers have a contract_type, while part-time drivers have working_hours.
- Some Vehicles are categorized by their type, such as Sedan, SUV or Electric. A vehicle can belong to only one type. Electric vehicles have BatteryCapacity, SUVs have cargoCapacity.
- A Part-time Driver can rent one Sedan at a time. When a Sedan is rented, the timeLimit is stored.
- A Full-time Driver can request repair from Staff on an SUV or an Electric. The issueDate is recorded when a repair request is made.
- A Staff will have StaffID, FullName, JoiningDate. The experience of a Staff can be calculated from JoiningDate.

Now, draw an **EER** (Extended Entity Relationship) diagram of this system. Write down all assumptions (if any).

Q3. [8 Marks] Construct a relational Schema by mapping the following EER diagram (Figure Q3). The following conditions must be fulfilled when constructing the schema:

- For mapping the 1:N or 1:1 relationships, you **MUST** use the foreign key approach.
- For mapping the superclasses/subclasses, **select** the applicable option between 8B or 8C and **explain** briefly why the unselected option is not applicable for the given EER.

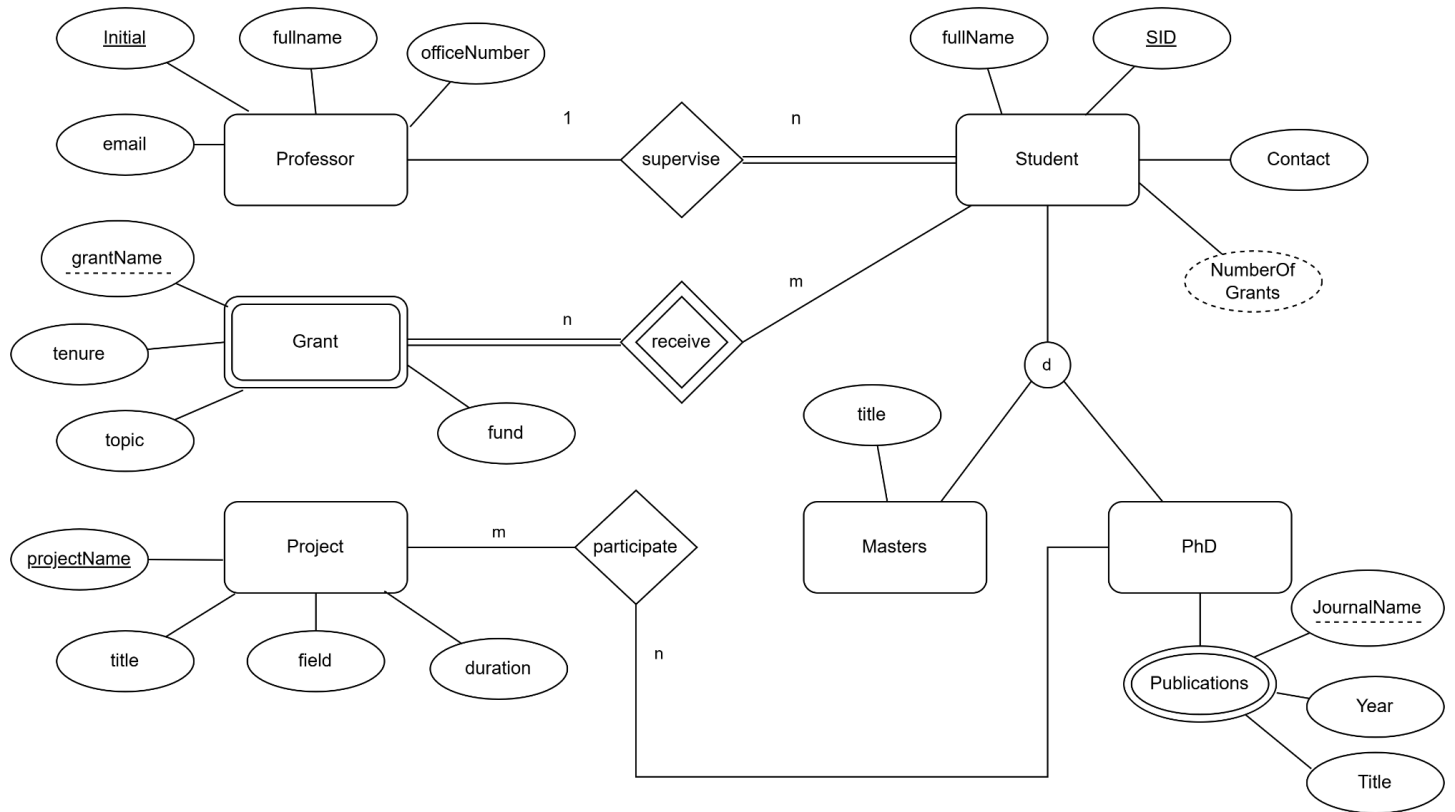


Figure: Q3